

Project Documentation

Import Data using Transform Maps (Spreadsheet)

The template is written for a MERN stack project. This project is built on ServiceNow, so each section is filled with the equivalent ServiceNow information, and sections that do not apply are marked Not applicable.

1. Introduction

- **Project Title:** Import Data using Transform Maps (Spreadsheet)
- **Team ID:** 6ab4d8aca238dc999eb5c33f
- **Team Members:** Barkavi K (add other members and roles here)
- **Date:** 03 October 2026

2. Project Overview

- **Purpose:** Import employee training records from a spreadsheet into ServiceNow quickly and accurately, link them to employees and protect them with role-based access.

Features:

- Custom table Employee Training Records with four fields
- Bulk upload through Load Data and a Transform Map
- Employee reference with Department dot-walk
- HR Manager role, application menu and module
- Read and Write ACLs verified by impersonation

3. Architecture

Template Item	This Project
Frontend	ServiceNow web UI: Load Data page, table list, form layout and application module (no React)
Backend	ServiceNow platform: import set staging table, Transform Map and ACL engine (no Node.js / Express)
Database	ServiceNow tables: Employee Training Records (Training Name, Completion Date, Status, Employee) and sys_user

4. Setup Instructions

- **Prerequisites:** ServiceNow instance with administrator access; Excel or Google Sheets; web browser.

Installation steps:

- Create the Employee Training Records table with four fields.
- Add In Progress and Completed to the Status choices.
- Prepare the spreadsheet with the same four columns.
- Upload it through Load Data.

- Create the Transform Map and run Transform.
- Create the HR Manager role and ACLs.

5. Folder Structure

Not applicable. ServiceNow configuration is stored in the instance, so there are no client and server folders. The configuration items are the table, Transform Map, role, application module and ACLs.

6. Running the Application

No npm commands are needed. Log in to the ServiceNow instance, open the HR Manager application module and open Employee Training Records. To import new data, go to Load Data, upload the file, then run Transform.

7. API Documentation

No custom API endpoints were created. Data is loaded through the built-in Load Data and Import Set process.

Item	Description
Method	Upload via Load Data (UI)
Parameters	Spreadsheet file with four columns
Response	Import set rows staged, then records inserted on Transform

8. Authentication

Users sign in with their ServiceNow credentials and sessions are managed by the platform. Authorisation uses the HR Manager role with Read and Write ACLs on the table. ACLs are created after elevating to the security_admin role. No separate tokens are used.

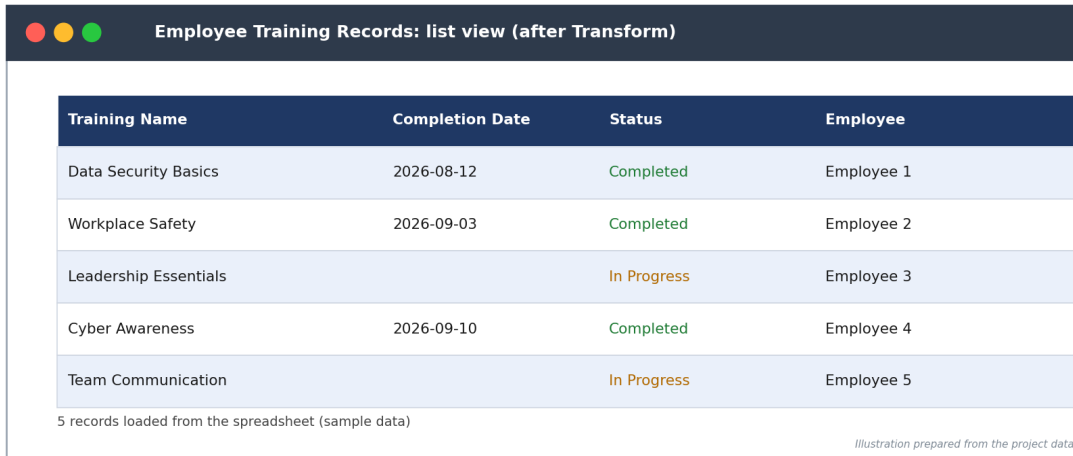
9. User Interface

Figure 1: Employee Training Records form with dot-walked Department (illustration)

10. Testing

Manual functional and security testing was carried out with user impersonation. Ten test cases covered table setup, import, mapping, employee linking and access control, and all ten passed.

11. Screenshots or Demo



Training Name	Completion Date	Status	Employee
Data Security Basics	2026-08-12	Completed	Employee 1
Workplace Safety	2026-09-03	Completed	Employee 2
Leadership Essentials		In Progress	Employee 3
Cyber Awareness	2026-09-10	Completed	Employee 4
Team Communication		In Progress	Employee 5

5 records loaded from the spreadsheet (sample data)

Illustration prepared from the project data

Figure 2: Imported records in the list view (illustration)

12. Known Issues

- Spreadsheet columns must match the expected structure.
- Employees must exist in sys_user for the reference to resolve.
- Duplicate uploads are not blocked unless a coalesce field is set.

13. Future Enhancements

- Scheduled imports
- Coalesce fields to avoid duplicates
- Department-wise reports and dashboards
- Reminders for trainings still In Progress